CONTOSO HEALTH SYSTEM

Clinical Knowledge Assistant → Scenario Brief & Knowledge Base Summary

LC 2 Project Module Resource | Composite / illustrative organization for coursework use | v1.0

1. Organization Profile

Organization	Contoso Health System (CHS)
Footprint	12 hospitals, 85 outpatient clinics, 5 states (Great Lakes region)
Workforce	45,000 employees, including approx. 6,800 credentialed prescribers and 11,200 clinical staff
EHR platform	Enterprise EHR (Epic-class) with a unified clinical data warehouse; regional variation in local order sets
Regulatory scope	HIPAA Privacy & Security Rules, 42 CFR Part 2 (behavioral health subset), state pharmacy board rules (5 states), FDA labeling for drug-interaction content
Program sponsor	Clinical Informatics, in partnership with Pharmacy & Therapeutics Committee and Compliance
Delivery team	GenAIOps engineering, Clinical SME panel (rotating), Pharmacy safety reviewers, Compliance/Privacy office, Audit owner

2. Program Context

CHS is deploying an enterprise clinical knowledge assistant to reduce time spent searching protocols, cut avoidable drug-interaction errors, standardize patient education materials, and accelerate administrative lookups. The assistant is retrieval-augmented: it answers from an approved, versioned knowledge base rather than open-domain generation, and every response must be traceable to a source passage and a prompt/index version pair.

The system launches into a regulated, multi-state clinical environment. Prior incidents in comparable health-system deployments have involved plausible-sounding but ungrounded protocol answers and stale drug-interaction guidance following formulary updates. CHS's leadership has therefore made auditability and risk-tiered review a launch condition, not a post-launch enhancement.

3. Use Cases and Risk Classification

Four use cases are in scope for the initial experimentation framework. Risk tier drives every governance, testing, and evaluation decision made downstream — the same tiers must be

used consistently across the PromptOps policy, the experiment matrix, and the evaluation gates.

Use Case	Risk Tier	Why	Primary Reviewer
Drug-interaction checks	Critical	Direct patient-safety impact; wrong or missed interaction can cause harm	Pharmacy SME + Clinical Safety Officer
Protocol lookup	Medium-High	Guides clinical action but typically reviewed by clinician before execution	Clinical SME (specialty-matched)
Patient education generation	Medium-High	Patient-facing language; low direct clinical action risk, high reputational/health-literacy risk	Clinical SME + Health Literacy reviewer
Administrative guidance	Low	Scheduling, billing, HR/policy questions; no direct clinical consequence	Departmental owner (spot-check only)

4. Knowledge Base Summary

Sources	Approved clinical protocols (internal), formulary + drug-interaction reference (licensed, updated monthly), patient education library (internal, literacy-graded), HR/administrative policy manuals
Volume	~14,000 protocol and reference documents; ~3,200 patient-education templates; ~900 administrative policy documents
Update cadence	Formulary/drug-interaction data: monthly (or emergency out-of-cycle for safety recalls). Protocols: quarterly review cycle. Admin policy: ad hoc
Indexing	Domain-segmented indices (pharmacy, clinical protocol, patient education, administrative) rather than one unified index, to allow independent freshness windows and independent rollback
PHI exposure	Knowledge base itself contains no PHI. Prompts must not include patient identifiers; query logs are subject to minimization and retention limits under the PromptOps authoring standards

5. Constraints the Framework Must Respect

- No PHI may appear in prompt text, logged queries, or evaluation datasets used for testing.
- Any change touching Critical-tier content (drug interactions) requires pharmacist SME sign-off before any production exposure, including silent/shadow mode.
- All five states' pharmacy board rules must be satisfiable by the most restrictive state, not the average.

- Every production-promoted change must be reconstructable from an audit trail: who proposed it, what evaluation it passed, who approved it, and when it went live.
- Rollback must be executable inside a defined SLA once a safety-relevant incident trigger fires.

6. What This Brief Is For

This brief supplies the scenario context referenced as [PLACEHOLDER] in the LC 2 Project Module. Use it as the shared source of truth when completing the PromptOps Governance Policy, RAG Experiment Matrix, Evaluation Framework, and Experimentation Workflow Diagram. Risk tiers, use-case names, and constraints defined here should be used verbatim across all four deliverables to keep the submission internally consistent.